

NGS file formats



Primary format for NGS data

FASTQ

- Format that associates sequences with quality value by nucleotide base

BAM

- Format for aligned and not aligned sequences
- Is binary (compressed)

SAM

- Format for aligned and not aligned sequences
- Text format (extended and human readable)

FASTA format

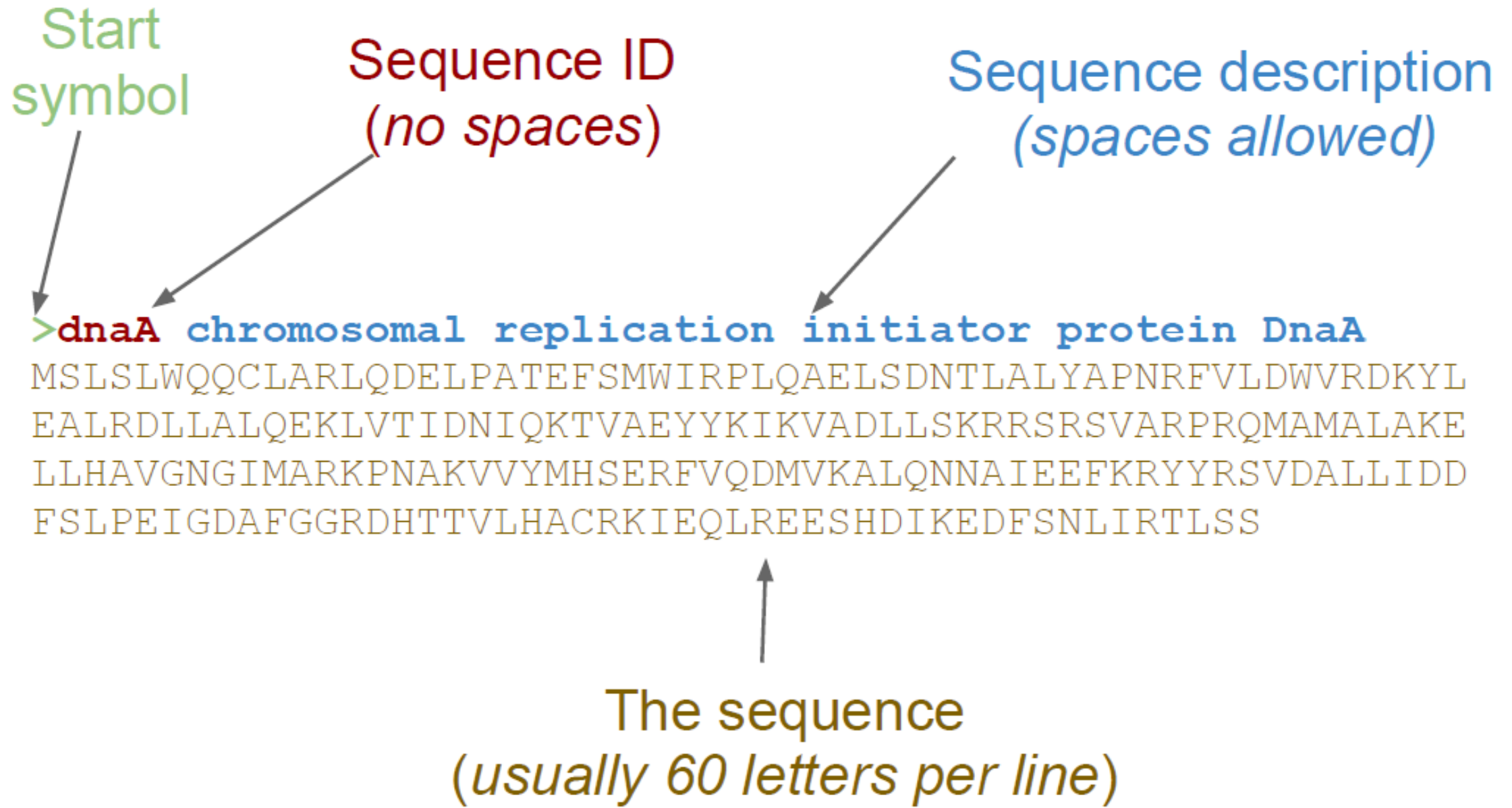
```
>dnaA chromosomal replication initiator protein DnaA
MSLSLWQQCLARLQDELPATEFSMWIRPLQAE LSDNTLALYAPNRFVLDWVRDKYL
EALRDLLALQEKLVTIDNIQKTVAEYYKIKVADLLSKRRSRSVARPRQMAMALAKE
LLHAVGNGIMARKPNAKVVYMHSE RFVQDMVKALQNN AIEEFKRY YRSVDALLIDD
FSLPEIGDAFGGRDHTTVLHACRKIEQLREESHDIKEDFSNLIRTLSS
```

Formato FASTA: Componentes

Start
symbol

Sequence ID
(no spaces)

Sequence description
(spaces allowed)



```
>dnaA chromosomal replication initiator protein DnaA
MSLSLWQQCLARLQDELPATEFSMWIRPLQAELSDNTLALYAPNRFVLDWVRDKYL
EALRDLLALQEKLVTIDNIQKTVAEYYKIKVADLLSKRRSRSVARPRQMAMALAKE
LLHAVGNGIMARKPNAKVVMHSEFVQDMVKALQNNAIEEFKRYYSVDALLIDD
FSLPEIGDAFGGRDHTTVLHACRKIEQLREESHDIKEDFSNLIRTLSS
```

The diagram illustrates the components of a FASTA format entry. It shows a sample entry with a start symbol, a sequence ID, a sequence description, and a multi-line sequence. Arrows point from the labels to the corresponding parts of the entry. The sequence is wrapped at 60 characters per line.

The sequence
(usually 60 letters per line)

Archivos Multi-fasta

- Are created concatenating individual FASTA sequences
- The symbol “>” separate the different sequences

```
>dnaA Chromosomal replication initiator protein DnaA
MSLSLWQQCLARLQDELPATEFSMWIRPLQAELSDNTLALYAPNRFVLDWVRDKYL
EALRDLLALQEKLVTIDNIQKTVAEYYKIKVADLLSKRRSRSVARPRQMAMALAKE
LLHAVGNGIMARKPNAKVVMHSEFVQDMVKALQNNALIEEFKRYRSVDALLIDD
FSLPEIGDAFGGRDHTTVLHACRKIEQLREESHDIKEDFSNLIRTLSS
```

```
>TetX tetanus toxin coding sequence
ATGGCGCGCCGGGACAGAAATGCCCTGCAGGAAGTTCTTCTGG
AAGACCTTCTCCTCCTGCAAATAAAACCTCACCCATGAATGC
CTCACGCAAGTTTAATTACAGACCTGAA
```

```
>CHP_3431 hypothetical protein
MTVLACRKIEQLREESHDIKEDFSNLIRTLSSMSLSLWQQCLARQDEL
SLPEIGDAFGGRDHTLQAELSDNTLALYAPNRFVLDWVRDKYLNINE
LVNAKVVY
```

FASTA Alphabets

DNA

- Standard: AGTC (4)
- Extended: adds N (unknown base)
- Full: adds RY MS WK VHDB (ambiguous bases)
 - R = A/G (puRine), Y=C/T (pYrimidine)

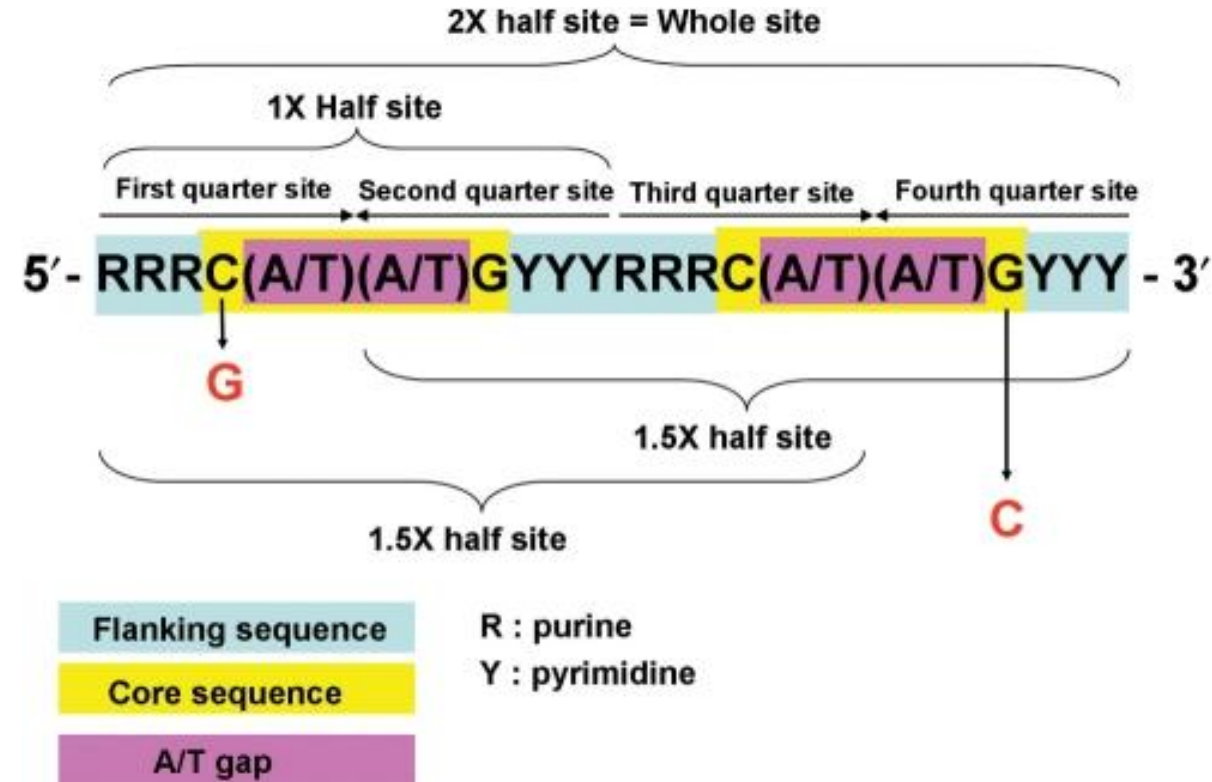
Protein

- Standard: ARNDC QEGHI LKMFP SUTWY V (21)
- Extended: adds X (unknown amino acid), * (term)
- Full: adds OBZJ
 - O = pyrrollysine, B = D/N, Z = Q/E, J = I/L

RNA

- replace T with U in DNA rules

Consensus Sequence

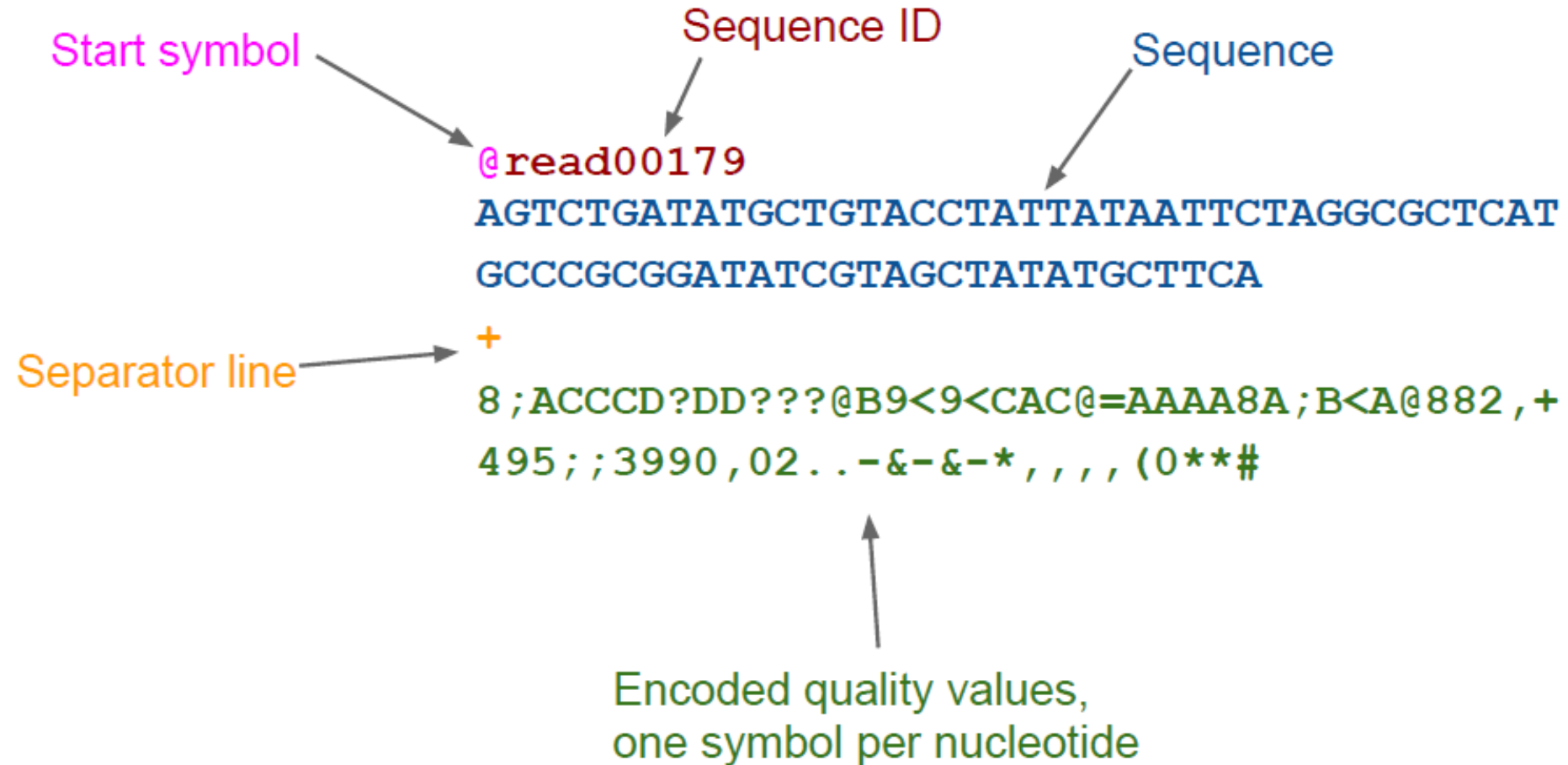


FASTA Extension

Suffix	Alphabet	Usage
.fasta .fa .fsa	Any	Generic
.fna	DNA	Specifically large genomic chunks eg. contigs, chromosomes
.faa	Protein	Peptide amino acid sequences
.ffn	DNA	Genomic coding regions (CDS)
.frn	DNA	Non-coding RNA (ncRNA) eg. tRNA, rRNA, siRNA, ...

Formato FASTQ

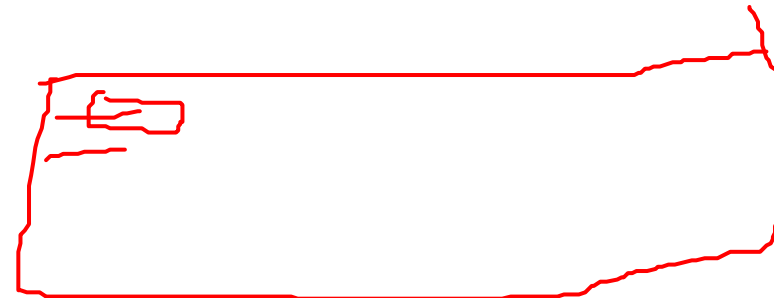
Is an extension of the FASTA format carrying quality values associated with each base



FASTQ format — Illumina sequence information

```
@M01478:6:000000000-A40C5:1:1101:16859:1439:1:1:N:0:1
ATCGTTTCGGAGCAAGGCAACTGTNTCAGGCACCATGAAGTTGAGCTATTCTACTGCGCCAACCTTTGCGAGATAAA
TCGTCNTGCCNTNNTTATCANC GTCAATTGGAANTCAGATGTGCCACCNNAAN
+
ABBBAABFB BBBBGE GGGGGGGHF#AAFF2AGFGHGH HHHHHHFHFFDGF GHHHHGHEGGGGCGGGGHFABEEGFF
HHEGHEGE#BBFG#?##???FFH#??FEFGHHEHHG#???FFEDGGGGFFHFH##??#
```

- Machine name
- Run number
- Flowcell ID
- Flowcell lane
- Tile in flowcell
- X-coordinate in tile
- Y-coordinate in tile
- Member of Pair (1/2)
- Read filtered? (Y/N)
- Control bits on (0 or even number)
- Index sequence used



SAM format

SAM refers to **S**equence **A**lignment/**M**ap format

Tab-delimited text formed by two sessions:

- Headers (start with @)
- Alignment (11 columns describing alignment features)

SAM format Example

For the following alignment:

```
Coord      12345678901234 5678901234567890123456789012345
ref         AGCATGTTAGATAA**GATAGCTGTGCTAGTAGGCAGTCAGCGCCAT

+r001/1      TTAGATAAAGGATA*CTG
+r002        aaaAGATAA*GGATA
+r003        gcctaAGCTAA
+r004                ATAGCT.....TCAGC
-r003                ttagctTAGGC
-r001/2                        CAGCGGCAT
```

This is the corresponding SAM:

```
@HD VN:1.6 SO:coordinate
@SQ SN:ref LN:45
r001    99 ref    7 30 8M2I4M1D3M = 37 39 TTAGATAAAGGATACTG *
r002     0 ref    9 30 3S6M1P1I4M * 0 0 AAAAGATAAGGATA *
r003     0 ref    9 30 5S6M          * 0 0 GCCTAAGCTAA * SA:Z:ref,29,-,6H5M,17,0;
r004     0 ref   16 30 6M14N5M       * 0 0 ATAGCTTCAGC *
r003  2064 ref   29 17 6H5M          * 0 0 TAGGC * SA:Z:ref,9,+,5S6M,30,1;
r001   147 ref   37 30 9M            = 7 -39 CAGCGGCAT * NM:i:1
```

Header lines

Describe the data origin, reference sequence, alignment method, etc

```
@HD VN:1.6 SO:coordinate
@SQ SN:ref LN:45
r001    99 ref    7 30 8M2I4M1D3M = 37   39 TTAGATAAAGGATACTG *
r002     0 ref    9 30 3S6M1P1I4M * 0    0 AAAAGATAAGGATA *
r003     0 ref    9 30 5S6M          * 0    0 GCCTAAGCTAA   * SA:Z:ref,29,-,6H5M,17,0;
r004     0 ref   16 30 6M14N5M      * 0    0 ATAGCTTCAGC   *
r003 2064 ref   29 17 6H5M          * 0    0 TAGGC        * SA:Z:ref,9,+,5S6M,30,1;
r001  147 ref   37 30 9M            = 7 -39 CAGCGGCAT   * NM:i:1
```

@HD: First line

VN= SAM Version

SO=Sequence order (coordinates or name)

@SQ: Reference sequence SN=Sequence name LN=Reference sequence length

SP=specie

@RG: Read information part

ID=Sequence Identifier CN=sequencing center SM=sample name

@PG: Programa

PN=Program name VN= Program Version

Para la explicación completa de los diferentes tipos de encabezado revise el documento:

<http://samtools.github.io/hts-specs/SAMv1.pdf>

Alignments are reported on 11 columns

```
@HD VN:1.6 SO:coordinate
```

```
@SQ SN:ref LN:45
```

```
r001 99 ref 7 30 8M2I4M1D3M = 37 39 TTAGATAAAGGATACTG *
r002 0 ref 9 30 3S6M1P1I4M * 0 0 AAAAGATAAGGATA *
r003 0 ref 9 30 5S6M * 0 0 GCCTAAGCTAA *
r004 0 ref 16 30 6M14N5M * 0 0 ATAGCTTCAGC *
r003 2064 ref 29 17 6H5M * 0 0 TAGGC *
r001 147 ref 37 30 9M = 7 -39 CAGCGGCAT *
```

Optional

```
SA:Z:ref,29,-,6H5M,17,0;
SA:Z:ref,9,+,5S6M,30,1;
NM:i:1
```

Col	Field	Type	Regexp/Range	Brief description
1	QNAME	String	[!-?A-~]{1,254}	Query template NAME
2	FLAG	Int	$[0, 2^{16} - 1]$	bitwise FLAG
3	RNAME	String	* [!-()+-<>-~] [!-~]*	Reference sequence NAME
4	POS	Int	$[0, 2^{31} - 1]$	1-based leftmost mapping POSition
5	MAPQ	Int	$[0, 2^8 - 1]$	MAPping Quality
6	CIGAR	String	* ([0-9]+[MIDNSHPX=])+	CIGAR string
7	RNEXT	String	* = [!-()+-<>-~] [!-~]*	Ref. name of the mate/next read
8	PNEXT	Int	$[0, 2^{31} - 1]$	Position of the mate/next read
9	TLEN	Int	$[-2^{31} + 1, 2^{31} - 1]$	observed Template LENgth
10	SEQ	String	* [A-Za-z=.]+	segment SEQUENCE
11	QUAL	String	[!-~]+	ASCII of Phred-scaled base QUALity+33

Second field: FLAGS describing alignment

```
@HD VN:1.6 SO:coordinate
@SQ SN:ref LN:45
r001 99 ref 7 30 8M2I4M1D3M = 37 39 TTAGATAAAGGATACTG *
r002 0 ref 9 30 3S6M1P1I4M * 0 0 AAAAGATAAGGATA *
r003 0 ref 9 30 5S6M * 0 0 GCCTAAGCTAA * SA:Z:ref,29,-,6H5M,17,0;
r004 0 ref 16 30 6M14N5M * 0 0 ATAGCTTCAGC *
r003 2064 ref 29 17 6H5M * 0 0 TAGGC * SA:Z:ref,9,+,5S6M,30,1;
r001 147 ref 37 30 9M = 7 -39 CAGCGGCAT * NM:i:1
```

Col	Field	Type	Regexp/Range	Brief description
1	QNAME	String	[!-?A-~]{1,254}	Query template NAME
2	FLAG	Int	[0, 2 ¹⁶ - 1]	bitwise FLAG
3	RNAME	String	* [!-()+-<>-~] [!-~]*	Reference sequence NAME
4	POS	Int	[0, 2 ³¹ - 1]	1-based leftmost mapping POSition
5	MAPQ	Int	[0, 2 ⁸ - 1]	MAPping Quality
6	CIGAR	String	* ([0-9]+[MIDNSHPX=])+	CIGAR string
7	RNEXT	String	* = [!-()+-<>-~] [!-~]*	Ref. name of the mate/next read
8	PNEXT	Int	[0, 2 ³¹ - 1]	Position of the mate/next read
9	TLEN	Int	[-2 ³¹ + 1, 2 ³¹ - 1]	observed Template LENgth
10	SEQ	String	* [A-Za-z=.]+	segment SEQUENCE
11	QUAL	String	[!-~]+	ASCII of Phred-scaled base QUALity+33

Explaining Alignment through a 12bit FLAG

Bit	Description
1	0x1 template having multiple segments in sequencing
2	0x2 each segment properly aligned according to the aligner
4	0x4 segment unmapped
8	0x8 next segment in the template unmapped
16	0x10 SEQ being reverse complemented
32	0x20 SEQ of the next segment in the template being reverse complemented
64	0x40 the first segment in the template
128	0x80 the last segment in the template
256	0x100 secondary alignment
512	0x200 not passing filters, such as platform/vendor quality controls
1024	0x400 PCR or optical duplicate
2048	0x800 supplementary alignment

Bits are combined, by seeming them, to describe alignment <https://broadinstitute.github.io/picard/explain-flags.html>

Sixth field: CIGAR reconstruct alignment

Draw the alignment!

```
@HD VN:1.6 SO:coordinate
```

```
@SQ SN:ref LN:45
```

```
r001 99 ref 7 30 8M2I4M1D3M = 37 39 TTAGATAAAGGATACTG *
r002 0 ref 9 30 3S6M1P1I4M * 0 0 AAAAGATAAGGATA *
r003 0 ref 9 30 5S6M * 0 0 GCCTAAGCTAA * SA:Z:ref,29,-,6H5M,17,0;
r004 0 ref 16 30 6M14N5M * 0 0 ATAGCTTCAGC *
r003 2064 ref 29 17 6H5M * 0 0 TAGGC * SA:Z:ref,9,+,5S6M,30,1;
r001 147 ref 37 30 9M = 7 -39 CAGCGGCAT * NM:i:1
```

```
Coord 12345678901234 5678901234567890123456789012345
ref AGCATGTTAGATAA**GATAGCTGTGCTAGTAGGCAGTCAGCGCCAT
```

```
+r001/1
+r002
+r003
+r004
-r003
-r001/2
```

- Sequences of numbers and bases associated with an operation
- Describing per base the alignment to the reference

Op	BAM	Description	Consumes query	Consumes reference
M	0	alignment match (can be a sequence match or mismatch)	yes	yes
I	1	insertion to the reference	yes	no
D	2	deletion from the reference	no	yes
N	3	skipped region from the reference	no	yes
S	4	soft clipping (clipped sequences present in SEQ)	yes	no
H	5	hard clipping (clipped sequences NOT present in SEQ)	no	no
P	6	padding (silent deletion from padded reference)	no	no
=	7	sequence match	yes	yes
X	8	sequence mismatch	yes	yes

Gene Annotation Formats

BED

- Format describing one feature (CDS, Exon, Intron, UTR, etc..) per line

GFF/GTF

- Format describing one feature (CDS, Exon, Intron, UTR, etc..) per line, often contains more information than bed format

BED format

- A tab-delimited file containing 3-12 columns of data
- It has 3 required fields that allow describing a feature:
 - **Chrom**: Name of chromosome or scaffold or any seq region use the suffix 'chr'
 - **ChromStart**: Start position of the feature
 - **ChromEnd**: End position of the feature

```
chr1 213941196 213942363
chr1 213942363 213943530
chr1 213943530 213944697
chr2 158364697 158365864
chr2 158365864 158367031
chr3 127477031 127478198
chr3 127478198 127479365
chr3 127479365 127480532
chr3 127480532 127481699
```

BED format

- Optional fields:
 - Name of the feature
 - Score (associated to track lines)
 - Strand (+ or -)
 - ThickStart: coordinate position which to start drawing a solid rectangle
 - ThickEnd: coordinate position which to stop drawing a solid rectangle
 - itemRgb: Colour values for the different tracks
 - blockCount: the number of sub-elements (e.g. exons) within the feature
 - blockSizes: the size of these sub-elements
 - blockStarts: the start coordinate of each sub-element

```
chr7 127471196 127472363 Pos1 0 + 127471196 127472363 255,0,0
chr7 127472363 127473530 Pos2 0 + 127472363 127473530 255,0,0
chr7 127473530 127474697 Pos3 0 + 127473530 127474697 255,0,0
chr7 127474697 127475864 Pos4 0 + 127474697 127475864 255,0,0
chr7 127475864 127477031 Neg1 0 - 127475864 127477031 0,0,255
chr7 127477031 127478198 Neg2 0 - 127477031 127478198 0,0,255
chr7 127478198 127479365 Neg3 0 - 127478198 127479365 0,0,255
chr7 127479365 127480532 Pos5 0 + 127479365 127480532 255,0,0
chr7 127480532 127481699 Neg4 0 - 127480532 127481699 0,0,255
```

GFF/GFT format

- A tab-delimited file containing 9 columns of data, plus optional files
- GFF and GFT are the same and come from (General Feature Format)
- Fields:
 - **seqname** - name of the chromosome or scaffold
 - **source** - name of the program that generated this feature or the data source
 - **feature** - feature type name, e.g. Gene, Variation, Similarity
 - **start** - Start position
 - **end** - End position
 - **score** - A floating-point value.
 - **strand** - defined as + (forward) or - (reverse).
 - **frame** - One of '0', '1' or '2'. '0' indicates that the first base of the feature is the first base of a codon, '1' that the second base is the first base of a codon, and so on..
 - **attribute** - A semicolon-separated list of tag-value pairs, providing additional information about each feature.

```
X      Ensembl Repeat  2419108 2419128 42      .      .      hid=trf; hstart=1; hend=21
X      Ensembl Repeat  2419108 2419410 2502    -      .      hid=AluSx; hstart=1; hend=303
X      Ensembl Repeat  2419108 2419128 0        .      .      hid=dust; hstart=2419108; hend=2419128
X      Ensembl Pred.trans.      2416676 2418760 450.19 -      2      genscan=GENSCAN00000019335
X      Ensembl Variation      2413425 2413425 .      +      .
X      Ensembl Variation      2413805 2413805 .      +      .
```